4190.428 001 Introduction to Machine Learning

Generalization and Regularization

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Training Error and Test Error

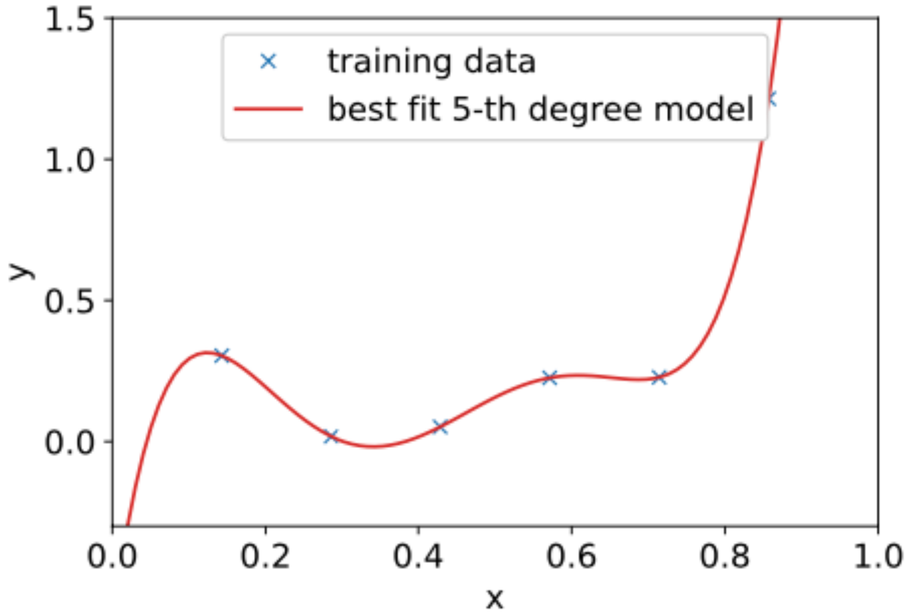
- **Training Error** : Loss function for **training data**
  - Example : (Training data is $\{x^{(i)}, y^{(i)}\}_{i=1}^{n}$)
  - $$J(\theta) = \frac{1}{n}\sum_{i=1}^{n}(y^{(i)} - h_\theta(x^{(i)}))^2$$

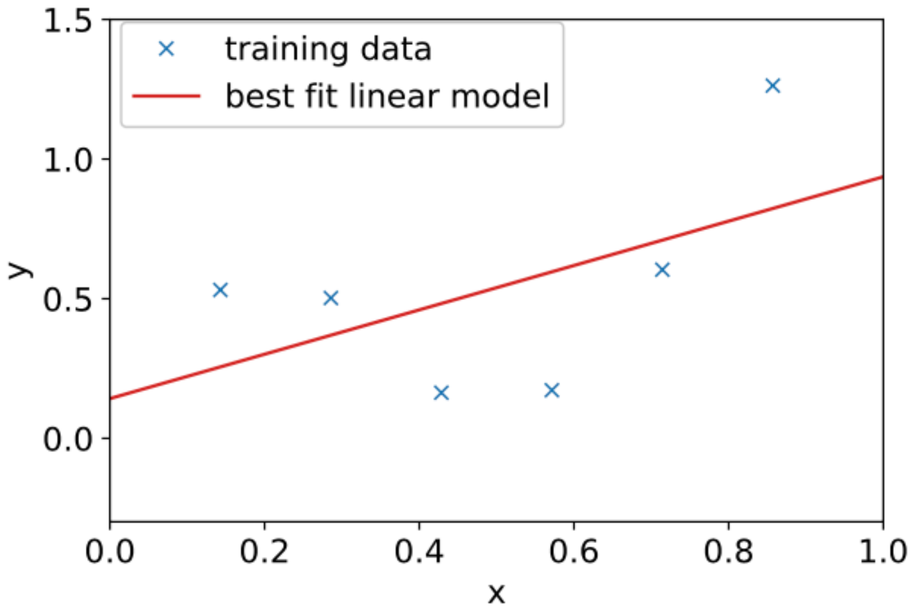
- **Test Error** : Loss on **unseen test examples**
  - Our ultimate goal is to minimize this Test Error!
  - Let the test data distribution be $\mathscr{D}$
  - $L(\theta) = \mathbb{E}_{(x,y)\sim\mathscr{D}}[(y - h_\theta(x))^2]$

- Difference between two errors are often written as **Generalization Gap**

Overfitting and Underfitting

- **Overfitting** : When model works (very) well on the training data, but doesn't work well on the test dataset
- **Underfitting** : When model poorly predicts on the training data (Not well trained)





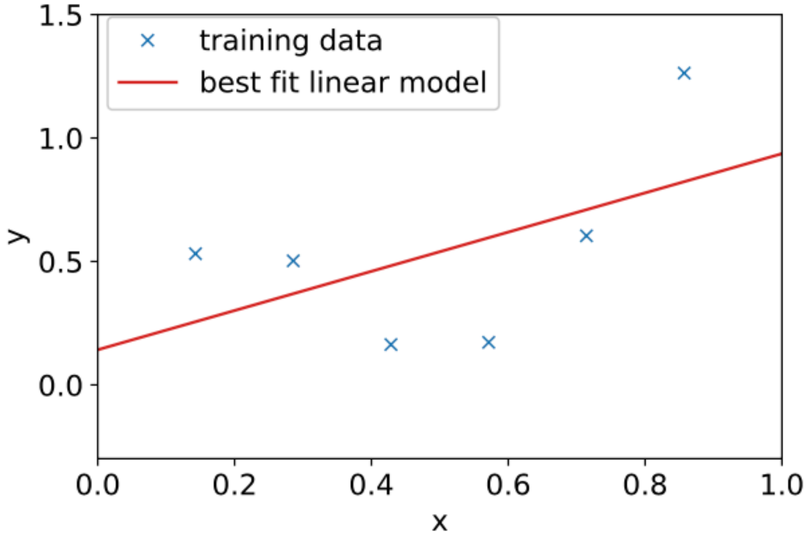
Bias and Variance

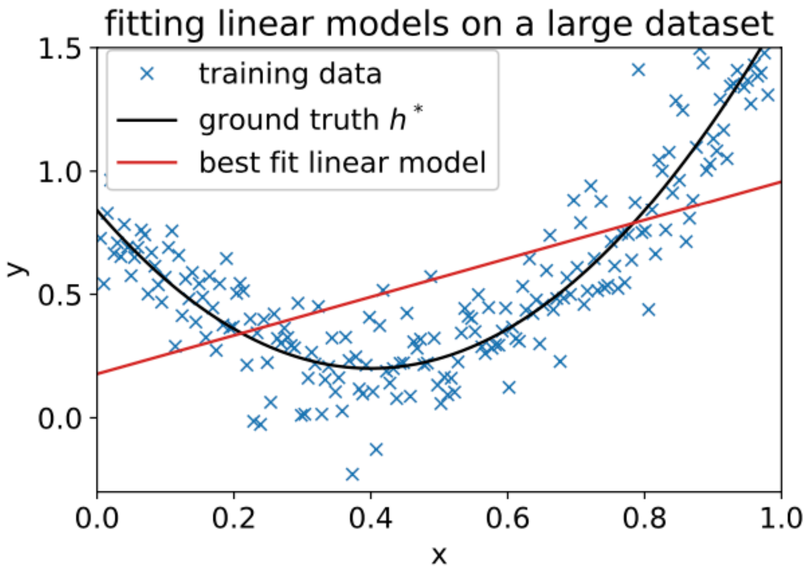
- Assume that we make a training dataset with following setting :
  - $y^{(i)} = h^*(x^{(i)}) + \xi^{(i)}$ where $h^*(\cdot)$ is a quadratic function
  - $\xi^{(i)} \sim N(0,\sigma^2)$ is an observation noise
  - Test example (x, y) is also drawn from $y = h^*(x) + \xi$

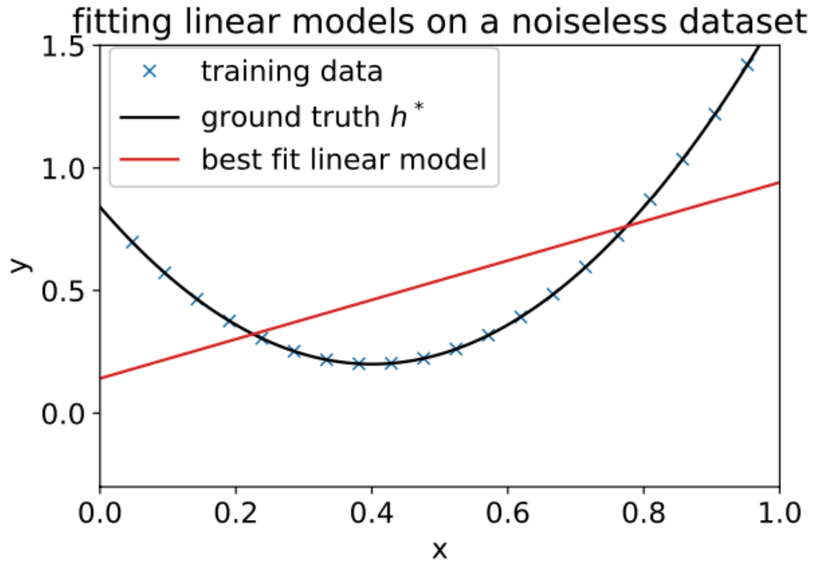
- We are going to use two models to recover $h^*(\cdot)$:
  - "Simple" Linear Model $y = \theta_0 + \theta_1 x$
  - "Complex" Polynomial Model $y = \theta_0 + \theta_1 x + \cdots + \theta_5 x^5$

Bias and Variance

- Linear Model **underfits** the data:
- This issue cannot be mitigated with more (even infinite) training examples
- Even we remove the observation noise, issue persists
- This is because of the inability of linear model family!

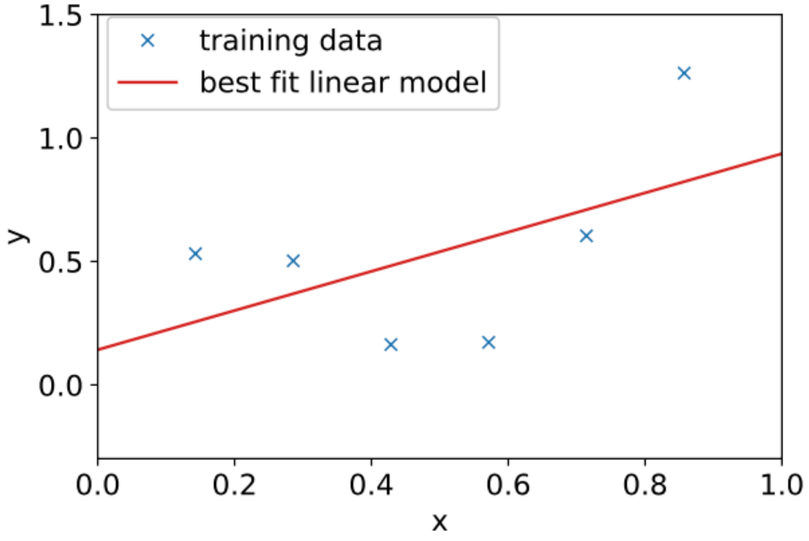


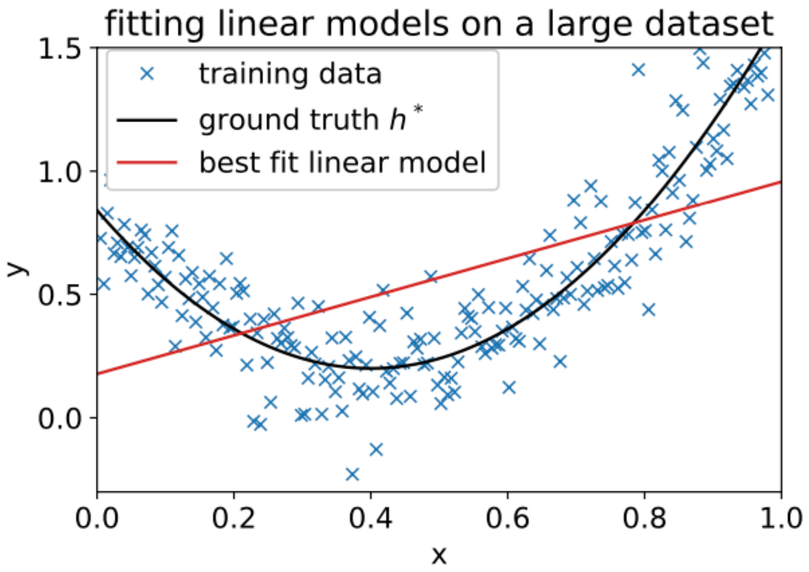


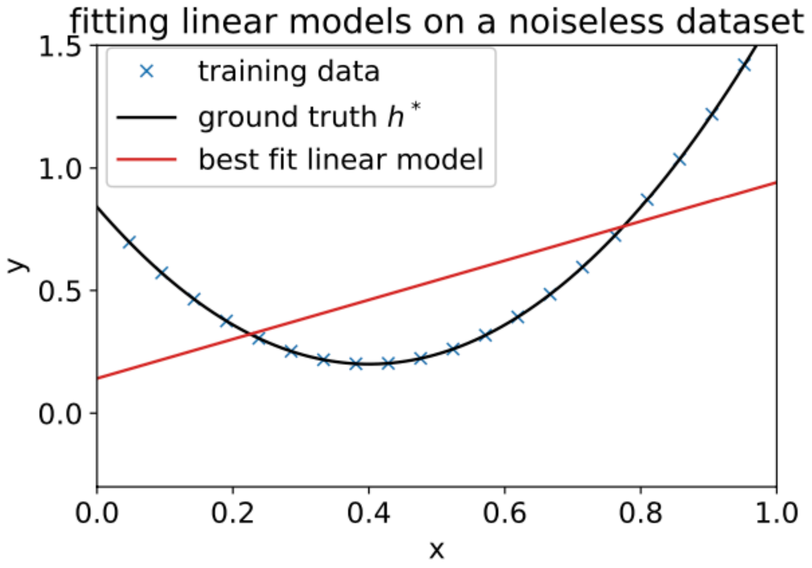


Bias and Variance

- We (informally) define **bias** of a model to be...
- The test error even if we were to fit it to a very large training dataset
- In this case, the linear model suffers from large bias!

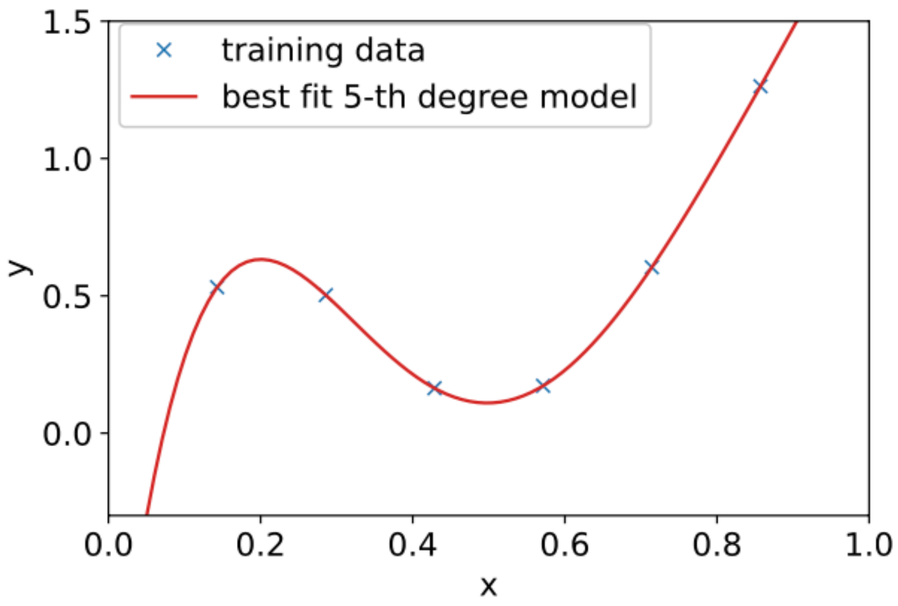


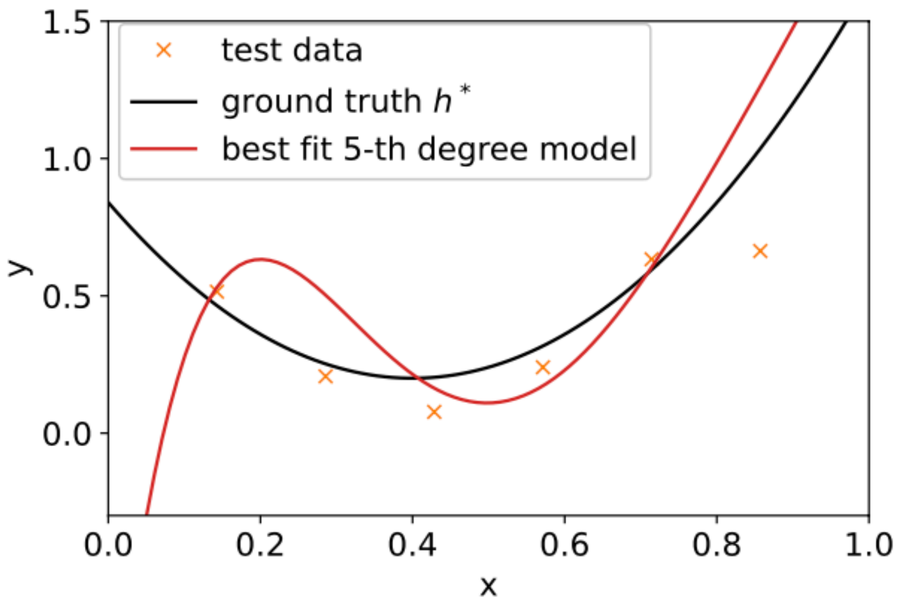




Bias and Variance

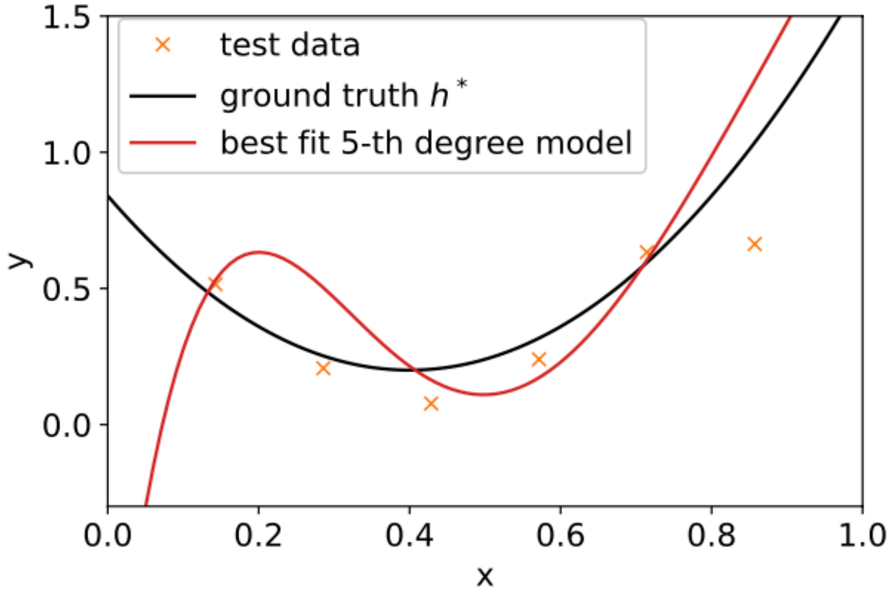
- 5th Polynomial Model **overfits** the data:
- It works very well on training examples, but works poorly on test examples
- Model learnt from the training set does not generalize well
- Test Error is high

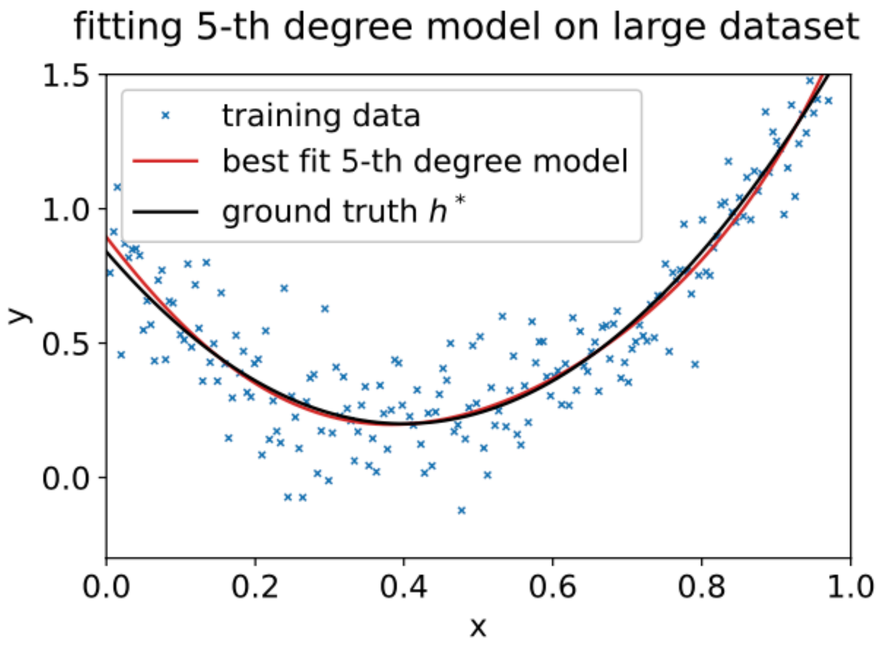




Bias and Variance

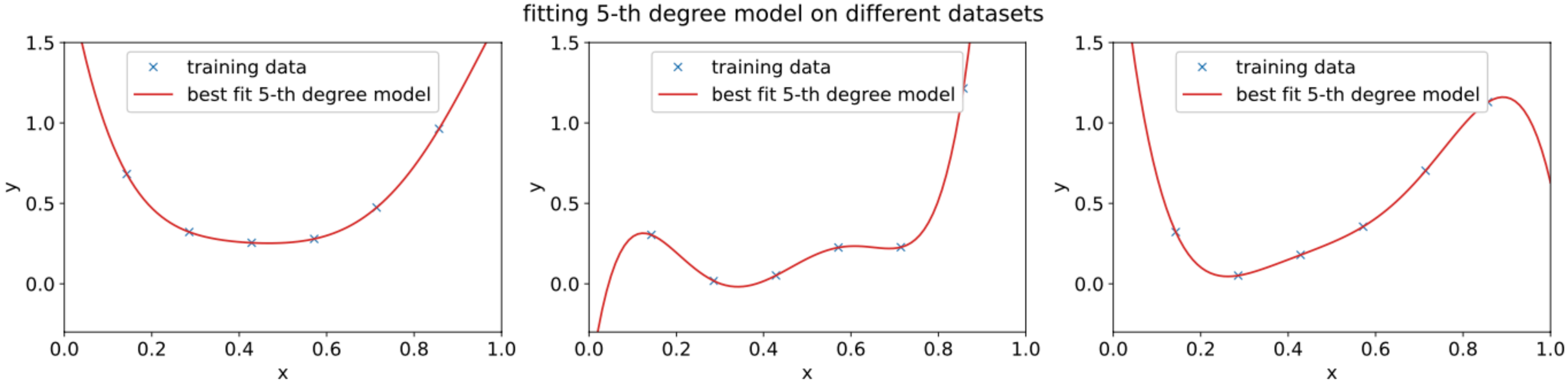
- But if we have bigger training dataset, result would be close to a original quadratic function (Since we can set $\theta_5 = \theta_4 = \theta_3 = 0$, while linear model can't)
- Thus its bias is small!





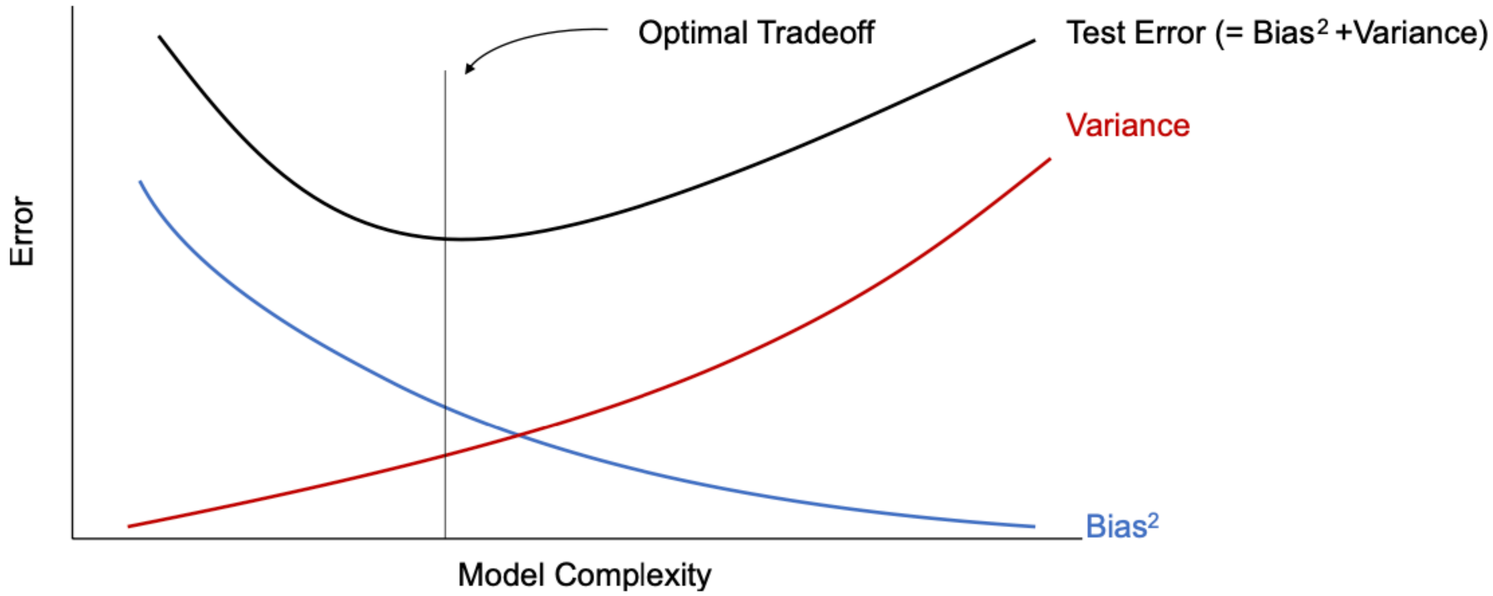
Bias and Variance

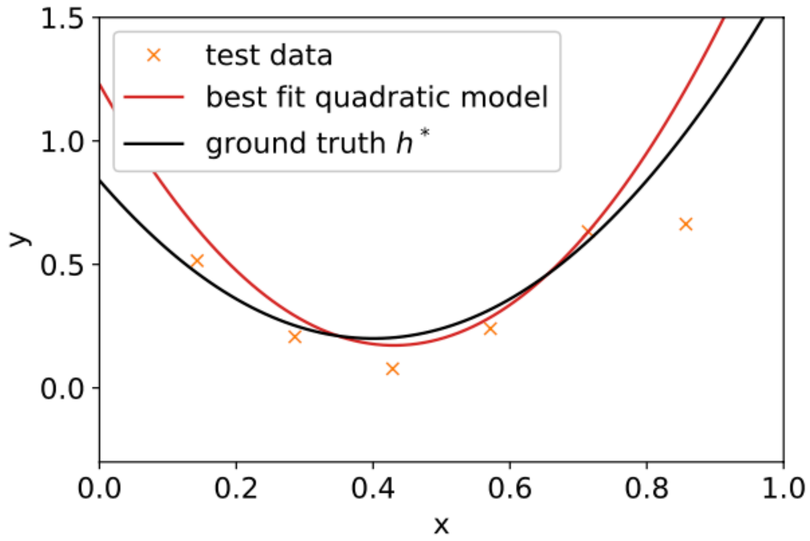
- When we have a small, finite training set, the resulting model is very different depending on the dataset
- We call this as a **variance**, and it indicates the amount of variations across models learnt on multiple different training datasets



Bias-Variance Tradeoff

- We often have a **tradeoff** between bias and variance
- For example, using a quadratic model is a optimal tradeoff





Bias-Variance Tradeoff

- Let $S = \{x^{(i)}, y^{(i)}\}_{i=1}^{n}$ be a training dataset, such that $y^{(i)} = h^*(x^{(i)}) + \xi^{(i)}$
- $\xi^{(i)} \in N(0,\sigma^2)$, and we will train a model $\hat{h}_S$
- Using the same method, we take a test example (x, y) and define the expected test error $\text{MSE}(x) = \mathbb{E}_{S,\xi}[(y - h_S(x))^2]$

$$
\begin{aligned}
\text{MSE}(x) &= \mathbb{E}[(y - h_S(x))^2] \\
&= \mathbb{E}[(\xi + (h^*(x) - h_S(x)))^2] \\
&= \mathbb{E}[\xi^2] + \mathbb{E}[(h^*(x) - h_S(x))^2] \text{ since } \mathbb{E}[\xi] = 0 \\
&= \sigma^2 + \mathbb{E}[(h^*(x) - h_S(x))^2]
\end{aligned}
$$

Bias-Variance Tradeoff

- Now, let $h_{avg}(x) = \mathbb{E}_S[h_S(x)]$ be an average model that is obtained by drawing an infinite number of datasets and averaging their predictions
- For many cases, h_{avg} is approximately equal to the model obtained by training on a single dataset with finite samples (like our intuition above)

Bias-Variance Tradeoff

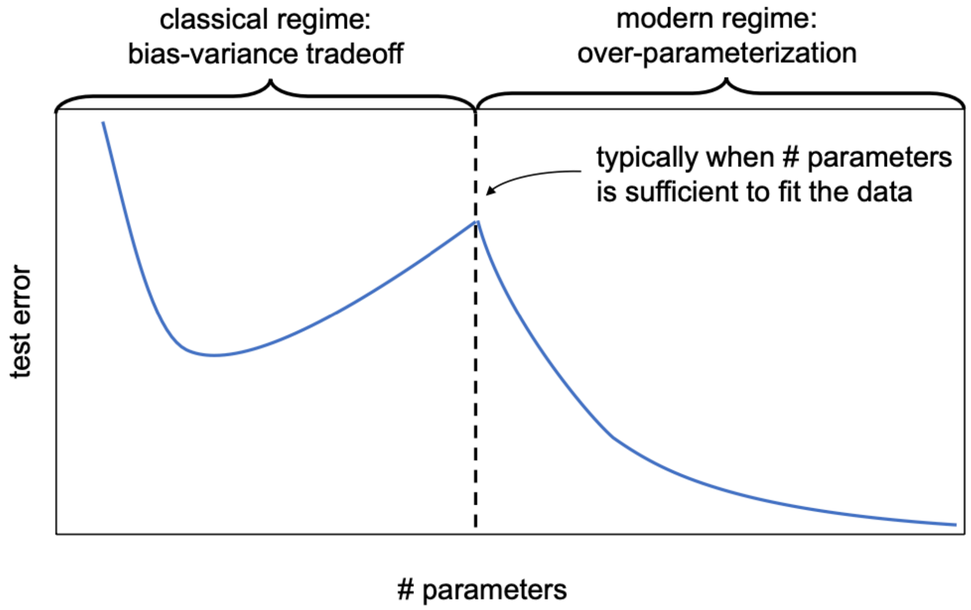
- As $c = h^*(x) - h_{avg}(x)$ is a constant, we can further modify the equation :

$$\begin{aligned}\text{MSE}(x) &= \sigma^2 + \mathbb{E}[(h^*(x) - h_S(x))^2] \\ &= \sigma^2 + (h^*(x) - h_{avg}(x))^2 + \mathbb{E}[(h_{avg} - h_S(x))^2] \\ &= \sigma^2 + (h^*(x) - h_{avg}(x))^2 + Var(h_S(x))\end{aligned}$$

- First term is a **unavoidable (noise) variance**, Second term is a **Bias (square)**, and the Third term is a **Variance**.

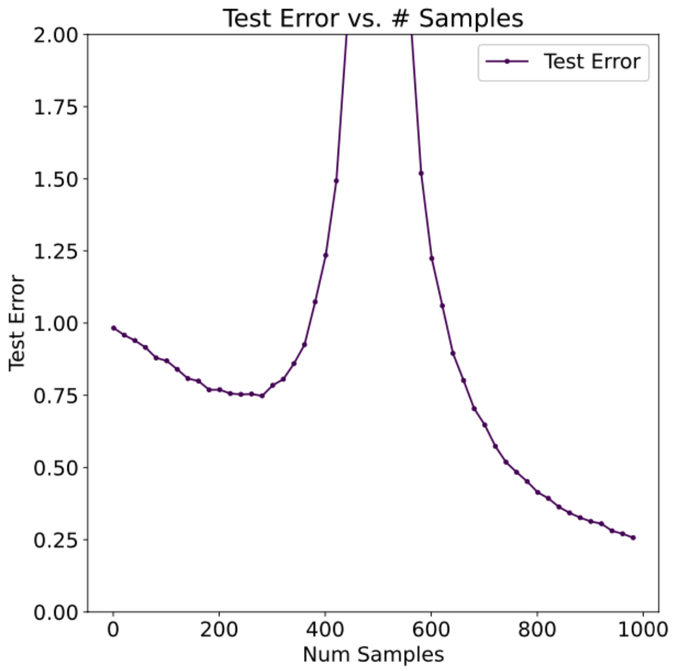
The Double Descent Phenomenon

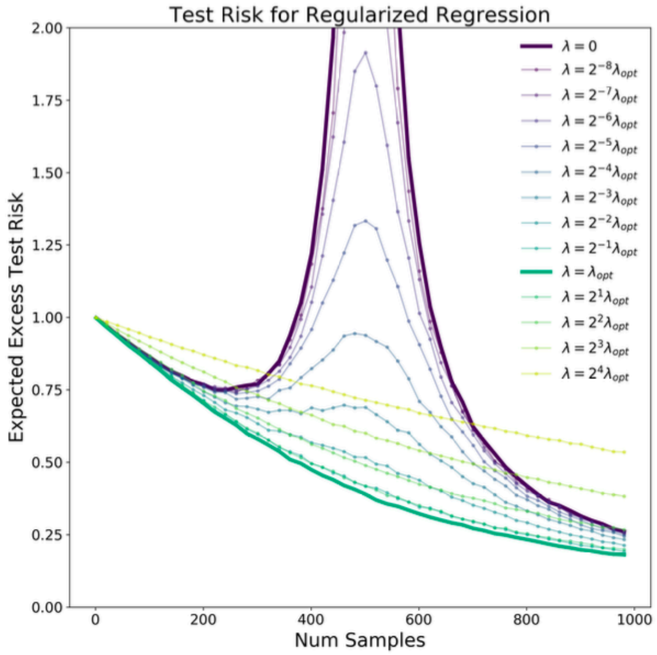
- In many cases in the practical area, actually we observe **the second descent** of the test error
- Test error has a peak when the number of parameters and samples are similar ($n \approx d$)
- In many cases, larger overparameterized models always lead to a better test performance!



The Double Descent Phenomenon

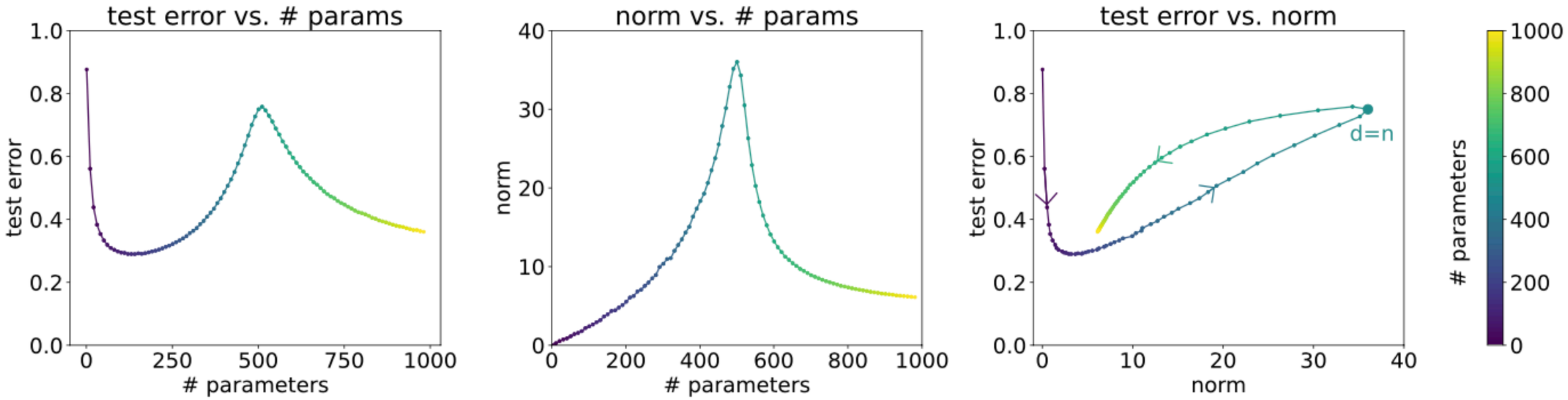
- We also has Sample-wise Double Descent phenomenon
- The test error is also peaked when $n \approx d$
- Later, we will learn about optimally-tuned regularization and use it to improve the test error in the $n \approx d$ regime (Picture on the Right)





The Double Descent Phenomenon

- If we take another measure for the model complexity like the norm of the models, the double descent phenomenon no longer occurs



QnA

- Any questions?
- Note: this lecture notes are largely adopted from
  - MIT 6.036(2019 Fall), Stanford CS229(2023-2024 Summer)